What is the cost of converting to Vegetarianism?

I. Introduction

With rising food costs, environmental factors and increased social pressure due to the awareness of environmental factors, more and more people may be wondering what they may gain or lose by converting to vegetarianism. The cost of converting one's diet may be measured in various ways: most obviously the change in money spent on groceries, but also in the opportunity cost of diversity in what one eats, the time one may have to give up in order to learn new methods of cooking, or how much time one spends preparing food. This report seeks to answer the question of what the cost is to convert to vegetarianism, by collecting the necessary data in order to make the required analysis. The data collection is driven by web scraping, through which food item prices are determined by scraping PnP's online grocery store, and recipes for common meals are found scraping RecipeTinEats' top most popular vegetarian dishes and top most popular dishes overall. The recipes are collected in order to give a view of what people are typically eating within either a vegetarian diet or non-vegetarian diet. The data scraped regarding food items is restructured to include an item description, price, quantity and metric of quantity. The data regarding recipes is restructured to contain information pertaining to the recipe's ingredients, cuisine origins, the courses the recipe can be served for as well as total servings and preparation time. Analysis of this data leads one to conclude that switching to vegetarianism has a far lower cost than one might assume, with similar levels of diversity and a slight reduction in cost for the average serving and a reduction in preparation time on average when cooking vegetarian meals.

II. Implementation

The following section details the web scraping implementation using `Scrapy`, the method by which the structured data is persisted, and the methods used to do analysis on the data collected.

A. Scrapy

I used `Scrapy` in order to scrape the web to collect the data I desired. In total I used three `Spiders`, two types of custom `Items`, and ten different `Pipelines`, of which different pipelines are enabled for different `Spiders`. I found no need to set a download delay, however I set it to 3 to be polite.

B. Database

Once the data is cleaned and structured in the `Scrapy` pipelines, the data is then stored in a `MySQL` database with the following schema:

TABLE I
Database tables and fields

Table Name	Fields
Products	Title, Price, Metric, Quantity, Barcode
Recipes	ID, Title, PrepTime, Servings, Vegetarian
RecipeIngredients	RecipeId, Description, Quantity, Metric
RecipeOrigins	RecipeId, Origin
RecipeCourses	RecipeId, Course

C. Analysis

In order to analyse the data collected, the following Python packages are utilised within a Jupyter Notebook: `mysql-connector`, `pandas`, `numpy`, `matplotlib` and `seaborn`. Most manipulation of data is done using `Dataframes` rather than native `SQL`, and all graphs are generated using `seaborn`.

In order to determine prices for recipes using food product pricing, I need to be able to match ingredients to products, and to convert quantities of ingredients and products. To match ingredients and products, I use a fuzzy string matcher. I find it works well in general, however, an incorrect match between something like 'almond milk' and 'whole almonds' can lead to unfortunate pricing jumps. To convert quantities and metrics, I use a variety of methods, some precise and others approximations. Some metrics can be converted exactly, such as liquids within the metric system. However, when measurements for solids are given in metrics such as teaspoons or cups, I make an approximate guess of what the weight would be. This was set up by inspection, by manually combing through ingredients and their respective product matches, and tweaking my conversion function until I was satisfied with the results.

III. Crawl Process

The following section details how the data was scraped and cleaned for both the food items and the recipes.

A. Food products

The food products spider, `PnpSpider`, has seven starting URLs, which cover the seven main categories of food products on PnP's online grocery store. Categories for non-food items were not included in the starting URLs, however any non-food items may have been scraped incidentally if incorrectly included within the website's own categories.

Each starting URL contains a page displaying 18 products, with pagination links appearing at the end of the product display. Every product on the page is parsed, and a pagination link to the next page of products is followed, until all products

in the category have been processed. For each product the title, price and product information are parsed into a `FoodItem`, which also includes initially blank fields for metric, quantity and barcode.

`FoodItems` are filtered and cleaned using pipelines. Firstly, blank and incorrectly scraped items are discarded. Then, any product with a title that has already been processed in the current session is discarded. The price information scraped is then cleaned by removing any HTML tags, the currency indicator, and converting the String to a Float. Note that the data scraped from PnP is very badly formatted, with large amounts of unrelated text appearing in a single tag along with important text, often requiring regex to clean the data. Following this, a two-step process is followed to extract the quantity and metric from the title of the `FoodItem`. The last processing step extracts the barcode from the large, free-text product information field. Finally, the item is inserted into the database given that the barcode is not already present.

B. Recipes

The recipes are scraped using two near identical spiders, `VegetarianSpider` and `PopularSpider`. They are kept separate as it is the easiest way to tag the category under which recipes fall without having to scrape that designation. Beyond this categorisation, the spiders are identical, and will be discussed as a unit. Each recipe spider has a single starting link, for the listing of most popular vegetarian recipes, or most popular recipes overall. Each of these landing pages contains 20 recipes each, with pagination links following the recipes. Each recipe on the landing page is parsed, after which a pagination link to the next page is followed. This is done until all recipes within the category are parsed. Each recipe is parsed into a `RecipeItem`, which includes fields for the recipe title, the preparation time, total servings, the cuisine origins, the meal courses, the diet designation and the recipe ingredients.

`RecipeItems` are filtered and cleaned using the following pipelines. Titles are cleaned to remove text not describing the recipe itself, then cuisine origins and courses are split into lists. Recipe ingredients are split into a list and each parsed to remove or convert imperial measurements for metric ones, as well as separating the name of the ingredient, the quantity and the metric. Finally, recipes are inserted into the database given that the recipe title is not yet present.

IV. Data Scraped

The following section details the quantity and quality of the data scraped, described in Section III.

A. PnP

It is my belief that I have scraped every product under the categories I scraped. I make this assumption based on the experience of scraping PnP's website, and the amount of products I have scraped. My impression of scraping PnP was that the website and service were thrown together very quickly in order to get to market as fast as possible, to gain market share within the already competitive grocery delivery sphere. I say this due to the unstructured nature of the HTML of the website, and the lack of protection against scraping. While scraping PnP, I never encountered any anti-scraping measures. I played around with scraping delays and found that even if I set no delay, the website would not block me, even after making 1000s of requests in seconds. In comparison, I played around with scraping Checkers and Woolworths before I decided to scrape PnP, and found myself blocked on Checkers by anti-botting techniques, and on Woolworths the content is loaded dynamically and would require Selenium or something similar to scrape. To test my theory, I rescraped the PnP data multiple times and got very consistent numbers of products. Therefore at 2462 products, I believe I have everything available by default within the categories I scraped. I will note that the PnP near where I live is not a very nice one, which likely influences the products listed for my IP.

B. RecipeTinEats

While my experience scraping RecipeTinEats was far more pleasant, the consistent use of measuring solids with liquid measurements was a great annoyance. There is also the assumption made that all recipes listed are for meals, which is not true. There are meals which describe how to make a sauce, or a dip, or something else which does not represent the cost to feed oneself. However, the consistency of the website justifies not excluding such recipes, as they appear under both vegetarian recipes and popular recipes. In total I scraped 157 vegetarian recipes and 182 popular recipes. In order to have equal representation from the two categories, a sample of 150 recipes is chosen at random every time a graph is created.

V. Results

This section details the data analysis and the conclusions drawn from the analysis in order to answer the questions posed.

A. What is the cost to diversity if one converts to vegetarianism?

The cost of diversity can be measured according to the variety one can enjoy within one's diet. Variety can be measured in the number of different cuisines one can enjoy, as well as the extent to which one's meals vary. To compare these, we look at the diversity of cuisines and the diversity of meal courses.

With reference to Figure 1, we can see that there is a larger diversity in the cuisines enjoyed by vegetarians than by that of non-vegetarians, as the vegetarian recipes have both a higher number of unique cuisines as well as having more cuisines unique to the category.

Looking at Figure 2, it can be seen that as a vegetarian one can expect Western food to take up a larger part of one's diet than before, and a reduction in cuisines frequently enjoyed by non-vegetarians such as Asian, Chinese and Italian food. Combining what can be seen in this graph with Figure 1, one can make the conclusion that while vegetarian food falls into more categories in terms of cuisines, non-vegetarian food is

Fig. 1. A comparison of the frequency and exclusivity of different cuisine origins between the most popular recipes and the most popular vegetarian recipes

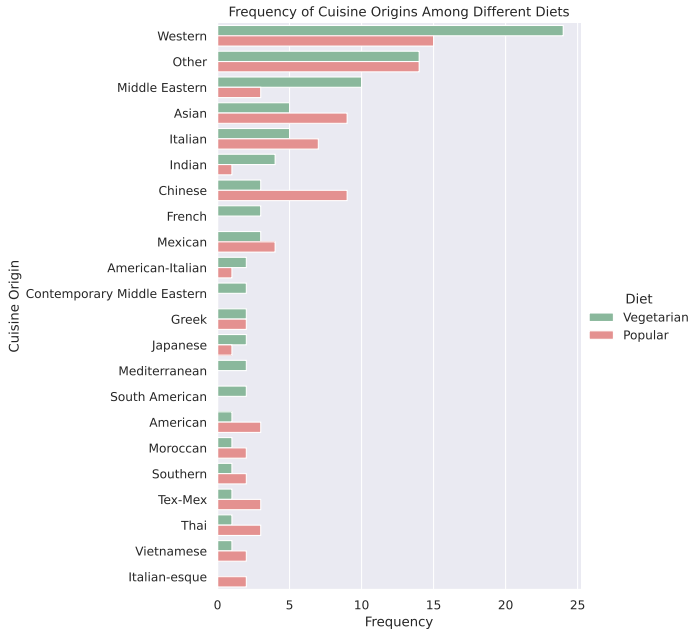

Fig. 2. The frequency of recipes being described as a specific cuisine origin for the most popular recipes and the most popular vegetarian recipes

more evenly distributed in terms of cuisines. Vegetarians may enjoy diversity overall, but the majority of their consumption will be of Western food. This suggests that vegetarianism may grant one more diversity at a the cost of not being able to specialise in a specific cuisine other than Western food, as there are not enough recipes within each other cuisine to meet that requirement.

Finally, looking at Figure 3, there is no notable lack of type of meal course for vegetarians. While differently distributed, for both dietary choices there are ample recipes which can be used for various different occasions and whims, and both diets enjoy a variety of choices.

In conclusion, the diversity cost of converting to vegetarianism is very low, and would not impact one's life dramatically.

B. What is the cost of one's time if one converts to vegetarianism?

The amount of time spent preparing meals is a appropriate indicator of the general effort added to one's life based off of one's diet. While time spent preparing food can often be offset through financial means, we wish to answer how reasonable it is for the average person to convert to vegetarianism. Looking at Figure 4, it can be seen that the deviation in preparation time for vegetarian meals is smaller than that of non-vegetarian meals, and that vegetarian meals typically take a shorter time in order to prepare a single serving of a meal. One may conclude that by switching to vegetarianism, one will not increase the amount of time one spends cooking. One should acknowledge the initial time spent to learn new recipes in order to fullfil a vegetarian diet, however in the long term one will not be spending more time.

C. What is the monetary cost of converting to vegetarianism?

Many might assume, that by converting to vegetarianism, one would spend less on groceries as one is not purchasing meat, which is often the most expensive portion of one's food bill. However, one might also realise that meat alternatives are typically expensive, especially in areas with a small vegetarian and vegan population. Looking at Figure 5 it can be seen that there is little difference in cost per serving between vegetarian meals and non-vegetarian meals. While the distribution of the cost per serving differs significantly above the $R100.00$ per serving mark, the distribution under that cost is very similar. One can conclude that this means that the barrier to entry to vegetarianism is very low, as one can eat cheaply while following both diets.

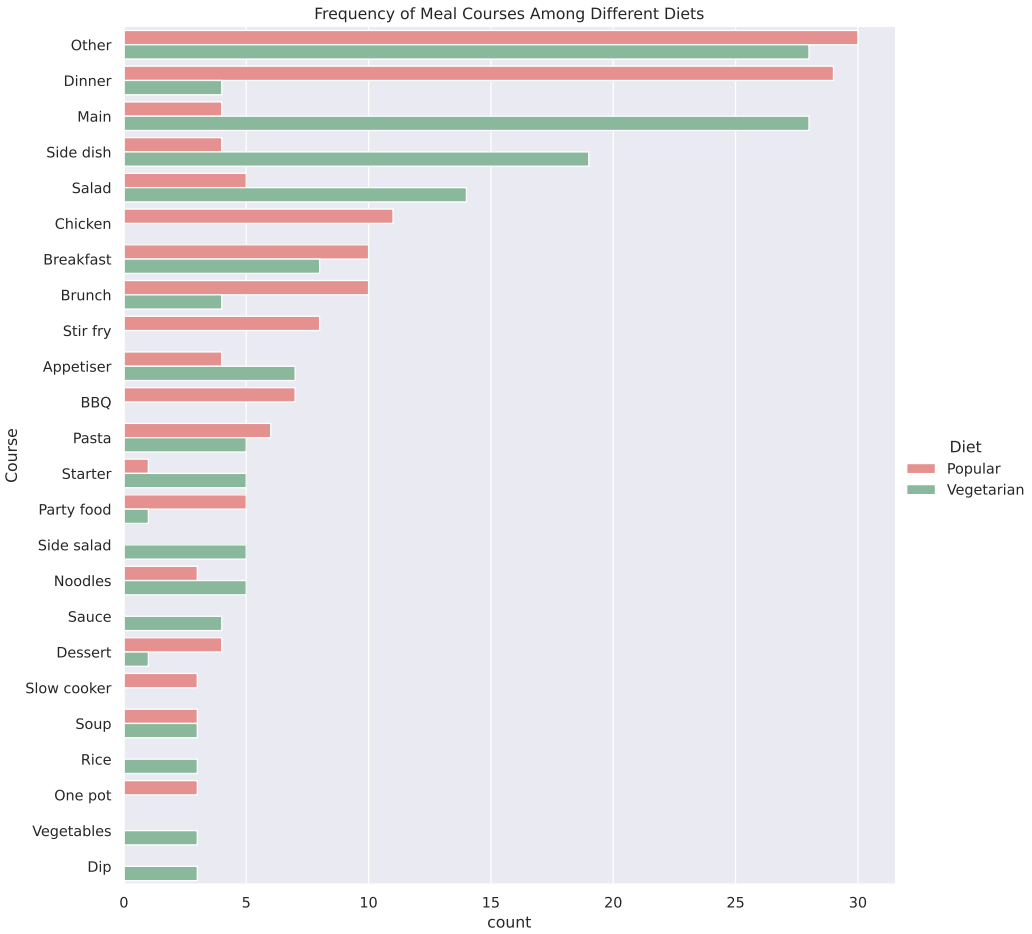

Fig. 3. The frequency of recipes being suitable to be served as a specific meal course for the most popular recipes and the most popular vegetarian recipes

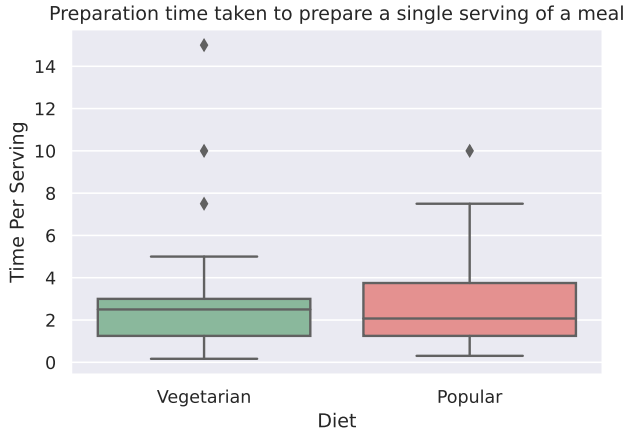

Fig. 4. The ratio between the time spent preparing a meal and the servings that meal provides, contrasted between two different diets

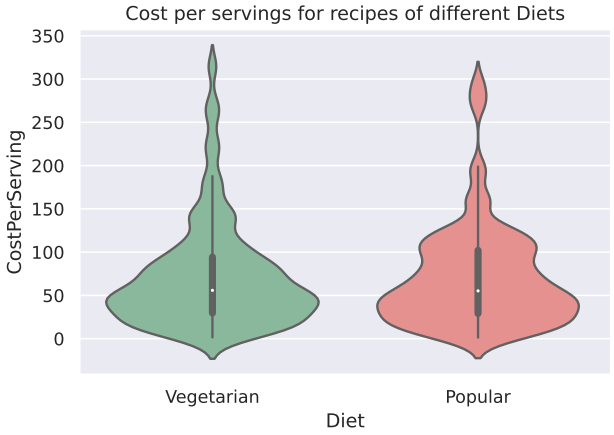

Fig. 5. The ratio between the total cost of a recipe and the number of servings that it provides, contrasted between two different diets

D. Conclusion

After inspection of the data collected, one can conclude that the cost to convert to vegetarianism is very low. While one's diet will change in terms of the cuisines consumed and the type of meals one will prepare, the time spent cooking and cost of feeding oneself propose no challenge to the average person wishing to convert to vegetarianism.